

Micro Energy Harvesting 2026**Exercise 2: Photovoltaics****Task 1: Irradiance**

On a cloudless summer day ($I_D = 850 \text{ W/m}^2$) the sun is at an angle of 45° towards the horizon.

- a) What is the optimal orientation of the solar module and what is the irradiance?

optimal alignment perpendicular to the angle of incident solar radiation, therefore at an angle β of 45°

In this case $S_i = I_{\text{Direkt}}$

$$\alpha = 45^\circ$$

$$\beta = 45^\circ$$

$$S_m = S_i \cdot \sin(\alpha + \beta) = 850 \cdot \sin(90^\circ) \text{ W/m}^2 = 850 \text{ W/m}^2$$

$$S_G = S_m \cdot 1,1 = 935 \text{ W/m}^2$$

Factor 1.1 is to be considered rather empirical, therefore no precise treatment of diffuse irradiation

- b) What is the irradiance at an angle of the solar module of 15° towards the horizon?

$$\beta = 15^\circ$$

$$S_m = S_i \cdot \sin(\alpha + \beta) = 850 \cdot \sin(60^\circ) \text{ W/m}^2 = 736 \text{ W/m}^2$$

$$S_G = S_m \cdot 1,1 = 810 \text{ W/m}^2$$

Now a winter day is considered: $I_D = I_{\text{Diff}} = 300 \text{ W/m}^2$. The sun is at an angle of 23° towards the horizon.

What is the irradiance for a solar module under the two angles of attack used above?

$$\alpha = 23^\circ$$

$$\beta = 15^\circ$$

$$S_m = S_i \cdot \sin(\alpha + \beta) = 185 \text{ W/m}^2$$

$$S_{m,\text{Diff}} = I_{\text{Diff}} \cdot \left(\frac{180^\circ - \beta}{180^\circ} \right) = 300 \frac{\text{W}}{\text{m}^2} \cdot \left(\frac{180^\circ - 15^\circ}{180^\circ} \right) = 275 \text{ W/m}^2$$

$$S_{\text{Global}} = S_m + S_{m,\text{Diff}} = 185 \text{ W/m}^2 + 275 \text{ W/m}^2 = \underline{\underline{460 \text{ W/m}^2}}$$

$$\alpha = 23^\circ$$

$$\beta = 45^\circ$$

$$S_m = S_i \cdot \sin(\alpha + \beta) = 278 \text{ W/m}^2$$

$$S_{m,Diff} = I_{Diff} \cdot \left(\frac{180^\circ - \beta}{180^\circ} \right) = 300 \frac{W}{m^2} \cdot \left(\frac{180^\circ - 45^\circ}{180^\circ} \right) = 225 \frac{W}{m^2}$$

$$S_{Global} = S_m + S_{m,Diff} = 278 \frac{W}{m^2} + 225 \frac{W}{m^2} = \underline{\underline{503 \frac{W}{m^2}}}$$

Task 2: Silicon solar cell

A silicon solar cell (band gap = 1.12 eV) with negligible parasitic resistance is uniformly illuminated by powerful sodium vapor lamps (total 1kW/m² at quasi-monochromatic 589 nm). The square cell has an edge length of 5 cm and an ideality factor of 1.1. The dark current I_0 is 1 pA at 300 K.

- a) What is the short-circuit current I_{SC} , open-circuit voltage V_{OC} , fill factor FF and efficiency η of the solar cell (assuming ideal QE and 100 % collection of all e⁻-h⁺ pairs).

$$E_G = 1,12 \text{ eV}$$

$$\lambda = 589 \text{ nm} = 5,89 \cdot 10^{-7} \text{ m}$$

$$I_G = 1000 \frac{W}{m^2}$$

$$A = (5 \text{ cm})^2 = 25 \text{ cm}^2 = 2,5 \cdot 10^{-3} \text{ m}^2$$

$$n = 1,1$$

$$I_0 = 1 \cdot 10^{-12} \text{ A}$$

$$T = 300 \text{ K}$$

$$QE = 1$$

$$SR(\lambda) = \frac{I_{SC}}{P_{in}} = \frac{q\lambda}{hc} QE \Rightarrow I_{SC} = \frac{q\lambda}{hc} QE \cdot P_{in}$$

$$P_{in} = I_G \cdot A = 1000 \frac{W}{m^2} \cdot 2,5 \cdot 10^{-3} \text{ m}^2 = 2,5 \text{ W}$$

$$I_{SC} = \frac{q\lambda}{hc} QE \cdot P_{in} = \frac{1,602176634 \cdot 10^{-19} \text{ C} \cdot 5,89 \cdot 10^{-7} \text{ m}}{6,62607015 \cdot 10^{-34} \frac{J}{s} \cdot 299792458 \frac{m}{s}} \cdot 1 \cdot 2,5 \text{ W} = \underline{\underline{1,19 \text{ A}}}$$

$$V_{OC} = \frac{nkT}{q} \ln \left(\frac{I_L}{I_0} + 1 \right) = \frac{1,1 \cdot 1,380649 \cdot 10^{-23} \cdot 300 \text{ K}}{1,602176634 \cdot 10^{-19} \text{ C}} \ln \left(\frac{1,19 \text{ A}}{1 \cdot 10^{-12} \text{ A}} + 1 \right) = \underline{\underline{0,79 \text{ V}}}$$

$$\begin{aligned} v_{oc} = \text{reduced voltage} &= \frac{q}{nkT} V_{OC} = \frac{1,602176634 \cdot 10^{-19} \text{ C}}{1,1 \cdot 1,380649 \cdot 10^{-23} \text{ J/K} \cdot 300 \text{ K}} \cdot 0,79 \text{ V} \\ &= 27,8 \frac{\text{C} \cdot \text{K} \cdot \text{V}}{\underbrace{\text{C} \cdot \text{V} \cdot \text{K}}_J} \end{aligned}$$

$$FF = \frac{v_{oc} - \ln(v_{oc} + 0.72)}{v_{oc} + 1} = \frac{27,8 - \ln(27,8 + 0.72)}{27,8 + 1} = \underline{\underline{0,849}}$$

$$\eta = \frac{P_{max}}{P_{in}} = \frac{V_{OC} \cdot I_{SC} \cdot FF}{P_{in}} = \frac{0,79 \text{ V} \cdot 1,19 \text{ A} \cdot 0,849}{2,5 \text{ W}} = \underline{31,9 \%}$$

- b) Corrosion on the front metal contacts leads to a series resistance R_S of 0.01Ω . How high is the proportion of power loss due to series resistance compared to the original cell?

$$R_S = 0,01 \Omega$$

$$R_{CH} = \frac{V_{OC}}{I_{SC}} = \frac{0,79 \text{ V}}{1,19 \text{ A}} = 0,666 \Omega$$

$$r_S = \text{Reduced resistance} = \frac{R_S}{R_{CH}} = \frac{0,01 \Omega}{0,666 \Omega} = 0,015$$

$$P_{max} = FF \cdot V_{OC} \cdot I_{SC} = 0,79 \text{ V} \cdot 1,19 \text{ A} \cdot 0,849 = 0,80 \text{ W}$$

$$FF_S = FF_0 \cdot (1 - 1.1r_S) + \frac{r_S^2}{5.4} = 0,849 \cdot (1 - 1.1 \cdot 0,015) + \frac{0,015^2}{5.4} = 0,835$$

$$P_{max}' = FF_S \cdot V_{OC} \cdot I_{SC} = 0,79 \text{ V} \cdot 1,19 \text{ A} \cdot 0,835 = \underline{0,78 \text{ W}}$$

$$\frac{P_{max}'}{P_{max}} = \frac{0,78 \text{ W}}{0,80 \text{ W}} = 98,4 \% \Rightarrow 1,6 \% \text{ Power lost}$$

Identical to: $\frac{FF_S}{FF}$

Task 3: Indoor Photovoltaics

A solar cell from Solems (type 07/055/020; www.solems.com) is to be used in a solar energy harvesting system.

- a) Determine and compare the FF and cell efficiency for the different lighting scenarios.

From the data sheet 07/055/020:

	200 lux	1000 lux	200 W/m ²	1000 W/m ²
V_{OC}	3,9 V	4,2 V	4,5 V	5 V
I_{SC}	15 μ A	70 μ A	2 mA	11 mA
P_{MP}	12 μ A @ 2,8 V	62 μ A @ 3V	1,8 mA @ 3,5 V	9,6 mA @ 3,7 V
$A = 11 \text{ cm}^2$				

Calculated

	200 lux	1000 lux	200 W/m²	1000 W/m²
$P_{MP} = I_{MP} \cdot V_{MP}$	0,03 mW	0,19 mW	6,30 mW	35,52 mW
$FF = \frac{P_{MP}}{V_{OC} \cdot I_{SC}}$	0,57	0,63	0,70	0,65
$P_{in} \left(100 \frac{Lm}{W} \right) in W$	0,0022	0,011	0,22	1,1
$\eta = \frac{P_{MP}}{P_{in}}$	1,53%	1,69%	2,86%	3,23%

b) Which battery system would you combine with this solar cell and how?

Only a Li-ion system with charge controller can be used sensibly here, whereby one must accept the resulting losses.

c) What else can the solar cell be used for besides supplying energy to the system?

Light sensor, as I_{SC} is proportional to the amount of light

Bonus Task 4: Position of the sun

On a clear November 12th, at 10:00 am local time in Toronto (Canada), someone wants to know the a) Hour Angle (HRA), b) Elevation Angle α and the Azimuth of the Sun.

Toronto: 43.66135°, -79.383087°, 76 m; UTC-5

<https://www.pveducation.org/pvcdrom/properties-of-sunlight/solar-time>

Hour Angle (HRA):

Local Standard Time Meridian LSTM = 15° · Δt_{UTC} = -75°

12. Nov. = 315+1 days since 1.1.

$$B = \frac{360}{365} (d - 81) = \frac{360}{365} (316 - 81) = 231,8$$

$$EoT = 9.87 \sin(2B) - 7.53 \cos(B) - 1.5 \sin(B) = 15,4 \text{ min}$$

$$TC = 4 \cdot (\text{Longitude} - LSTM) + EoT = 4 \cdot (-79,38^\circ - (-75^\circ)) + 15,4 \text{ min} = -2,1 \text{ min}$$

Online Calculator: -0,57 min

$$LST = LT + \frac{TC}{60} = 10 - \frac{13}{60} = 9,965 \Rightarrow \underline{9:57 \text{ Uhr}}$$

$$HRA = 15^\circ \cdot (LST - 12) = 15^\circ \cdot (9,965 - 12) = \underline{-30,5^\circ}$$

Elevation Angle α :

$$\delta = 23.45^\circ \cdot \sin \left[\frac{360}{365} \cdot (d - 81) \right] = 23.45^\circ \cdot \sin[230,8] = -18,42^\circ$$

$$\alpha = \sin^{-1}[\sin \delta \cdot \sin \varphi + \cos \delta \cdot \cos \varphi \cdot \cos (HRA)] = \arcsin[\sin(-18,42^\circ) \cdot \sin(43,66^\circ) + \cos(-18,42^\circ) \cdot \cos(43,66^\circ) \cdot \cos(-30,5^\circ)] = \underline{21,90^\circ}$$

Azimuth:

$$\text{Azimuth} = \cos^{-1} \left[\frac{\sin \delta \cdot \cos \varphi - \cos \delta \cdot \sin \varphi \cdot \cos (HRA)}{\cos \alpha} \right] =$$

$$\arccos \left[\frac{\sin(-18,42^\circ) \cdot \cos(43,66^\circ) - \cos(-18,42^\circ) \cdot \sin(43,66^\circ) \cdot \cos(-30,5^\circ)}{\cos(21,90^\circ)} \right] = \underline{153,5^\circ}$$